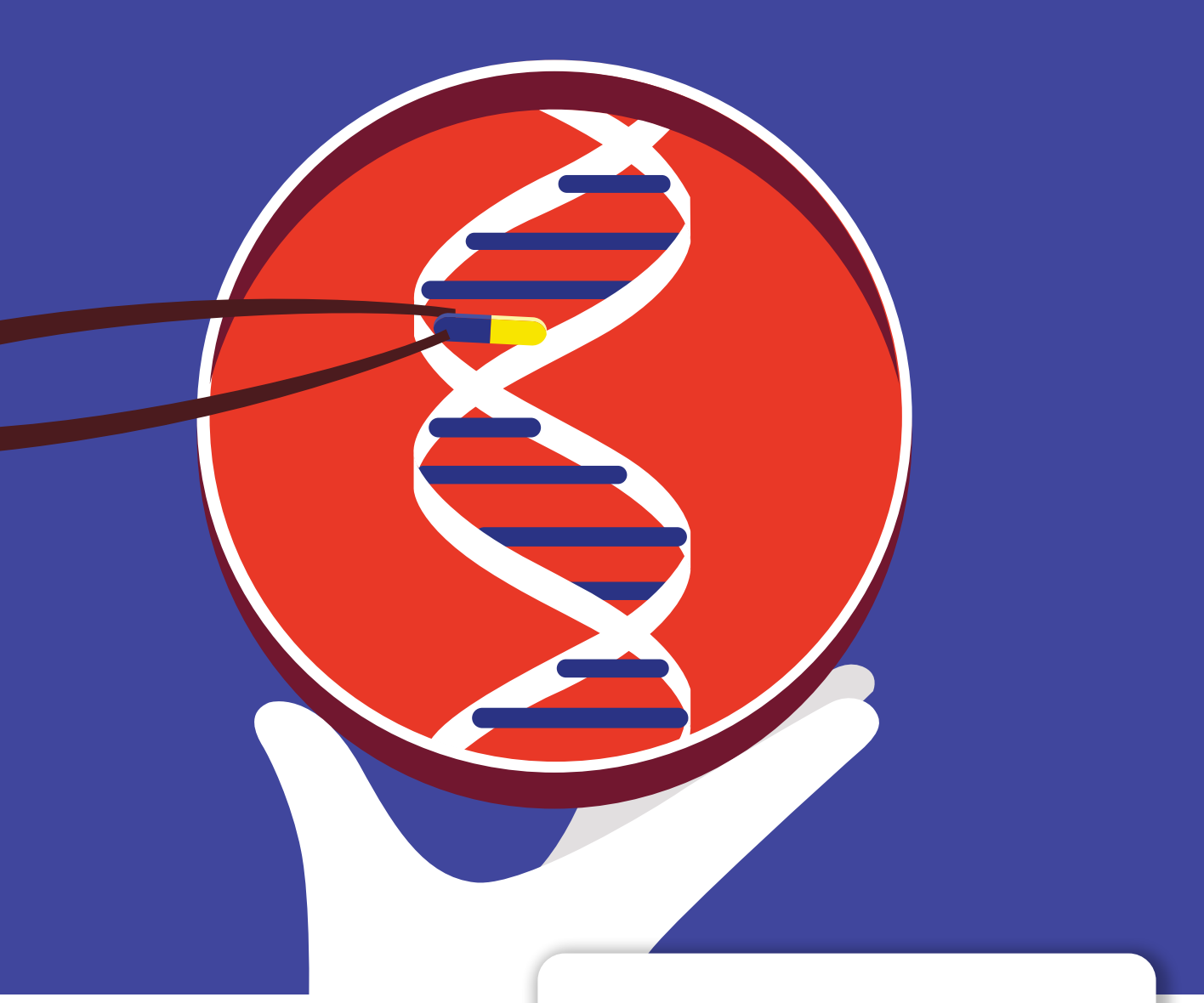


mutations are changes in the DNA sequence or genome—the linear arrangement of all gene sequences and other chemical bases—during cell division. If part of the DNA sequence is in the wrong place or incomplete, it might lead to genetic disorders and diseases. Genome sequencing, the laboratory process of determining the whole DNA sequence, can identify genetic causes of diseases.

"India has thousands of endogamous population groups with distinct genetic signatures," says Arjun Gupta, founder & CEO, Genetico, a research and diagnostics firm. "This means population-specific genetic variants and disease risks are different from Western populations." Government programmes like the Genome India Project have sequenced over 10,000 Indian genomes from 99 populations, creating a database meant to improve diagnosis, drug response prediction and treatment design. The Indian Cancer Genome Atlas is also trying to map mutation patterns specific to Indians.

How Precision Medicine Works

↘ GENETIC TESTING
Doctors collect a blood/ saliva or tumour tissue sample from the patient

↘ GENOME SEQUENCING
The patient's DNA is analysed to identify genetic mutations, disease markers or drug-response genes

↘ DATA ANALYSIS
Bioinformatics and AI systems analyse the genetic data, compare it with medical databases to identify actionable mutations

↘ TREATMENT MATCHING
Doctors select targeted drugs, immunotherapy, gene therapy or personalised treatment based on the patient's genetic profile

↘ MONITORING
Liquid biopsy/ genomic tests are used to monitor treatment response, detect relapse early and adjust treatment